

Chapter 16: Trigonometric Ratios in Triangles (Tỉ số Lượng giác trong Tam giác)

Chapter Overview

Trigonometric ratios extend beyond right triangles to apply to any triangle. The **Law of Sines** and **Law of Cosines** are powerful tools that allow us to solve triangles when we know certain combinations of sides and angles. These laws have practical applications in navigation, surveying, architecture, astronomy, and many fields requiring distance and angle measurements.

In this chapter, we will:

- Extend trigonometric ratios to angles from 0° to 180°
- Learn the Law of Cosines and its applications
- Learn the Law of Sines and its applications
- Study formulas for calculating triangle area
- Solve triangles given various combinations of sides and angles
- Apply these concepts to real-world problems in surveying and navigation

16.1 Bilingual Vocabulary (Từ vựng Song ngữ)

Core Concepts

English	IPA	Vietnamese	Example
Trigonometric ratio	/ˈtrɪɡənəˈmetrɪk ˈreɪʃiəʊ/	Tỉ số lượng giác	sin, cos, tan are trigonometric ratios
Sine	/saɪn/	Sin	$\sin A = \frac{a}{c}$ in a right triangle
Cosine	/ˈkoʊsaɪn/	Cosin	$\cos A = \frac{b}{c}$ in a right triangle
Tangent	/ˈtændʒənt/	Tang	$\tan A = \frac{a}{b}$ in a right triangle

Cotangent	/kou'tændʒənt/	Cotang	$\cot A = \frac{1}{\tan A}$
Acute angle	/ə'kju:t 'æŋgəl/	Góc nhọn	An angle less than 90°
Obtuse angle	/əb'tu:s 'æŋgəl/	Góc tù	An angle greater than 90° but less than 180°
Supplementary angles	/,sʌplɪ'mentəri 'æŋgəlz/	Hai góc bù nhau	Two angles that sum to 180°
Complementary angles	/,kɒmplɪ'mentəri 'æŋgəlz/	Hai góc phụ nhau	Two angles that sum to 90°

Laws and Formulas

English	IPA	Vietnamese	Formula
Law of Cosines	/lɔ: əv 'kouʒaɪnz/	Định lý cosin	$a^2 = b^2 + c^2 - 2bccos A$
Law of Sines	/lɔ: əv saɪnz/	Định lý sin	$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$
Area formula	/'eəriə 'fɔ:rmjələ/	Công thức diện tích	$S = \frac{1}{2}absin C$
Heron's formula	/'herənz 'fɔ:rmjələ/	Công thức Heron	$S = \sqrt{p(p-a)(p-b)(p-c)}$
Solving a triangle	/'sɒlvɪŋ ə 'traɪæŋgəl/	Giải tam giác	Finding all unknown sides and angles

Triangle Elements

English	IPA	Vietnamese	Notation
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Side	/saɪd/	Cạnh	a, b, c
Angle	/'æŋɡəl/	Góc	A, B, C
Opposite side	/'ɒpəzɪt saɪd/	Cạnh đối diện	Side a is opposite to angle A
Semi-perimeter	/,semi pə'rɪmɪtər/	Nửa chu vi	$p = \frac{a+b+c}{2}$
Circumradius	/'sɜːrkəmˌreɪdiəs/	Bán kính ngoại tiếp	R
Inradius	/'ɪnˌreɪdiəs/	Bán kính nội tiếp	r

16.2 Trigonometric Ratios for Angles 0° to 180° (Tỉ số Lượng giác cho Góc từ 0° đến 180°)

Extended Definition

In Grade 9, you learned trigonometric ratios for acute angles ($0^\circ < \alpha < 90^\circ$). Now we extend these definitions to any angle from 0° to 180° .

Definition: Trigonometric Ratios for Any Angle ($0^\circ \leq \alpha \leq 180^\circ$)

For any angle α ($0^\circ \leq \alpha \leq 180^\circ$), let $M(x_0, y_0)$ be the point on the unit semicircle such that $\angle XOM = \alpha$. Then:

- **$\sin \alpha$** = y_0 (the y-coordinate of M)
- **$\cos \alpha$** = x_0 (the x-coordinate of M)
- **$\tan \alpha$** = y_0/x_0 (when $\alpha \neq 90^\circ$)
- **$\cot \alpha$** = x_0/y_0 (when $\alpha \neq 0^\circ$ and $\alpha \neq 180^\circ$)

Special Values

α	0°	30°	45°	60°	90°	120°	135°	150°	180°
$\sin \alpha$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\cos \alpha$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{3}}{2}$	-1
$\tan \alpha$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	—	$-\sqrt{3}$	-1	$-\frac{\sqrt{3}}{3}$	0

Supplementary Angle Identities

Identities for Supplementary Angles

For any angle α ($0^\circ \leq \alpha \leq 180^\circ$):

- $\sin(180^\circ - \alpha) = \sin \alpha$
- $\cos(180^\circ - \alpha) = -\cos \alpha$
- $\tan(180^\circ - \alpha) = -\tan \alpha$
(when $\alpha \neq 90^\circ$)
- $\cot(180^\circ - \alpha) = -\cot \alpha$
(when $0^\circ < \alpha < 180^\circ$)

Key insight: Supplementary angles have equal sines but opposite cosines, tangents, and cotangents.

Example 16.1: Finding Trigonometric Values

Find all trigonometric values of 135° .

Solution:

Since $135^\circ = 180^\circ - 45^\circ$, we use the supplementary angle identities:

$$\sin 135^\circ = \sin(180^\circ - 45^\circ) = \sin 45^\circ = \frac{\sqrt{2}}{2}$$

$$\cos 135^\circ = \cos(180^\circ - 45^\circ) = -\cos 45^\circ = -\frac{\sqrt{2}}{2}$$

$$\tan 135^\circ = \tan(180^\circ - 45^\circ) = -\tan 45^\circ = -1$$

$$\cot 135^\circ = \cot(180^\circ - 45^\circ) = -\cot 45^\circ = -1$$

Answer:

- $\sin 135^\circ = \frac{\sqrt{2}}{2}$
 - $\cos 135^\circ = -\frac{\sqrt{2}}{2}$
 - $\tan 135^\circ = -1$
 - $\cot 135^\circ = -1$
-

16.3 The Law of Cosines (Định lý Cosin)

The Law of Cosines is a generalization of the Pythagorean theorem that works for **any triangle**, not just right triangles.

The Law of Cosines

In any triangle ABC with sides a , b , c opposite to angles A , B , C respectively:

$$a^2 = b^2 + c^2 - 2bccos A$$

$$b^2 = c^2 + a^2 - 2cacos B$$

$$c^2 = a^2 + b^2 - 2abcos C$$

Alternative forms (solving for the angle):

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos B = \frac{c^2 + a^2 - b^2}{2ca}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Note: When $C = 90^\circ$, $\cos C = 0$, and the Law of Cosines reduces to the Pythagorean theorem:

$$c^2 = a^2 + b^2$$

Example 16.2: Using the Law of Cosines (SAS Case)

In triangle ABC, $AB = 5$, $AC = 8$, and $\angle A = 120^\circ$. Find BC.

Solution:

We know two sides and the included angle (SAS case). Using the Law of Cosines:

$$BC^2 = AB^2 + AC^2 - 2 \cdot AB \cdot AC \cdot \cos A$$

$$BC^2 = 5^2 + 8^2 - 2(5)(8) \cos 120^\circ$$

$$BC^2 = 25 + 64 - 80 \cdot \left(-\frac{1}{2}\right)$$

$$BC^2 = 89 + 40 = 129$$

$$BC = \sqrt{129} \approx 11.36$$

Answer:

$$BC = \sqrt{129} \approx 11.36$$

Example 16.3: Using the Law of Cosines (SSS Case)

In triangle ABC, $a = 7$, $b = 8$, $c = 5$. Find angle C.

Solution:

We know all three sides (SSS case). Using the Law of Cosines:

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\cos C = \frac{7^2 + 8^2 - 5^2}{2(7)(8)}$$

$$\cos C = \frac{49 + 64 - 25}{112} = \frac{88}{112} = \frac{11}{14}$$

$$C = \arccos\left(\frac{11}{14}\right) \approx 38.21^\circ$$

Answer:

$$C \approx 38.21^\circ$$

16.4 The Law of Sines (Định lý Sin)

The Law of Sines relates the sides of a triangle to the sines of its angles.

The Law of Sines

In any triangle ABC with sides a , b , c opposite to angles A , B , C respectively, and circumradius R :

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

Useful forms:

- $a = 2R \sin A$

,

- $b = 2R \sin B$

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- $c = 2R \sin C$

- $\sin A = \frac{a}{2R}$

,

- $\sin B = \frac{b}{2R}$

,

$$\sin C = \frac{c}{2R}$$

$$\cdot \frac{a}{b} = \frac{\sin A}{\sin B}$$

Example 16.4: Using the Law of Sines (ASA Case)

In triangle ABC, $\angle A = 45^\circ$, $\angle B = 60^\circ$, and $a = 10$. Find side b .

Solution:

We know two angles and one side (ASA case). First, find angle C :

$$C = 180^\circ - A - B = 180^\circ - 45^\circ - 60^\circ = 75^\circ$$

Using the Law of Sines:

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

$$\frac{10}{\sin 45^\circ} = \frac{b}{\sin 60^\circ}$$

$$b = \frac{10 \cdot \sin 60^\circ}{\sin 45^\circ} = \frac{10 \cdot \frac{\sqrt{3}}{2}}{\frac{\sqrt{2}}{2}} = \frac{10\sqrt{3}}{\sqrt{2}} = \frac{10\sqrt{6}}{2} = 5\sqrt{6}$$

$$b \approx 12.25$$

Answer:

$$b = 5\sqrt{6} \approx 12.25$$

Example 16.5: Using the Law of Sines (AAS Case)

In triangle ABC, $\angle A = 50^\circ$, $\angle C = 70^\circ$, and $c = 8$. Find side a .

Solution:

We know two angles and a non-included side (AAS case).

Using the Law of Sines:

$$\frac{a}{\sin A} = \frac{c}{\sin C}$$

$$\frac{a}{\sin 50^\circ} = \frac{8}{\sin 70^\circ}$$

$$a = \frac{8 \cdot \sin 50^\circ}{\sin 70^\circ} = \frac{8 \cdot 0.766}{0.940} \approx 6.52$$

Answer:

$$a \approx 6.52$$

16.5 Area Formulas for Triangles (Công thức Diện tích Tam giác)

Formula 1: Using Two Sides and Included Angle

Area Formula (SAS)

$$S = \frac{1}{2}ab \sin C = \frac{1}{2}bc \sin A = \frac{1}{2}ca \sin B$$

This formula is useful when you know two sides and the included angle.

Formula 2: Using All Three Sides (Heron's Formula)

Heron's Formula

If

$$p = \frac{a+b+c}{2}$$

is the semi-perimeter, then:

$$S = \sqrt{p(p-a)(p-b)(p-c)}$$

This formula is useful when you know all three sides.

Formula 3: Using Circumradius

$$S = \frac{abc}{4R}$$

where R is the circumradius.

Example 16.6: Area Using Two Sides and Angle

In triangle ABC , $AB = 6$, $AC = 8$, and $\angle A = 30^\circ$. Find the area.

Solution:

Using the SAS area formula:

$$S = \frac{1}{2} \cdot AB \cdot AC \cdot \sin A$$

$$S = \frac{1}{2} \cdot 6 \cdot 8 \cdot \sin 30^\circ$$

$$S = \frac{1}{2} \cdot 6 \cdot 8 \cdot \frac{1}{2} = 12$$

Answer: The area is 12 square units.

Example 16.7: Area Using Heron's Formula

In triangle ABC , $a = 5$, $b = 6$, $c = 7$. Find the area.

Solution:

First, find the semi-perimeter:

$$p = \frac{a+b+c}{2} = \frac{5+6+7}{2} = 9$$

Using Heron's formula:

$$S = \sqrt{p(p-a)(p-b)(p-c)}$$

$$S = \sqrt{9(9-5)(9-6)(9-7)}$$

$$S = \sqrt{9 \cdot 4 \cdot 3 \cdot 2} = \sqrt{216} = 6\sqrt{6} \approx 14.70$$

Answer: The area is

$$6\sqrt{6} \approx 14.70$$

square units.

16.6 Solving Triangles (Giải Tam giác)

Solving a triangle means finding all unknown sides and angles when given certain information.

Cases for Solving Triangles

Case	Given	Use
SSS	Three sides	Law of Cosines to find angles
SAS	Two sides and included angle	Law of Cosines to find third side
ASA	Two angles and included side	Law of Sines
AAS	Two angles and non-included side	Law of Sines
SSA	Two sides and non-included angle	Ambiguous case (may have 0, 1, or 2 solutions)

Example 16.8: Solving a Triangle (SAS)

In triangle ABC, $a = 7$, $b = 10$, and $\angle C = 50^\circ$. Solve the triangle.

Solution:

Step 1: Find side c using the Law of Cosines:

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$c^2 = 7^2 + 10^2 - 2(7)(10) \cos 50^\circ$$

$$c^2 = 49 + 100 - 140(0.643) = 149 - 90.02 = 58.98$$

$$c \approx 7.68$$

Step 2: Find angle A using the Law of Sines:

$$\frac{a}{\sin A} = \frac{c}{\sin C}$$

$$\sin A = \frac{a \sin C}{c} = \frac{7 \cdot \sin 50^\circ}{7.68} = \frac{7 \cdot 0.766}{7.68} \approx 0.698$$

$$A \approx 44.3^\circ$$

Step 3: Find angle B:

$$B = 180^\circ - A - C = 180^\circ - 44.3^\circ - 50^\circ = 85.7^\circ$$

Answer:

- $c \approx 7.68$
 - $A \approx 44.3^\circ$
 - $B \approx 85.7^\circ$
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16.7 Applications (Ứng dụng)

Application 1: Surveying and Distance Measurement

Problem: A surveyor needs to find the distance across a lake. From point A on one shore, she measures the distance to point B on the same shore as 200 m. She measures $\angle BAC = 65^\circ$ and $\angle ABC = 48^\circ$, where C is a point on the opposite shore. Find the distance AC.

Solution:

First, find angle C:

$$C = 180^\circ - 65^\circ - 48^\circ = 67^\circ$$

Using the Law of Sines:

$$\frac{AC}{\sin B} = \frac{AB}{\sin C}$$

$$AC = \frac{AB \cdot \sin B}{\sin C} = \frac{200 \cdot \sin 48^\circ}{\sin 67^\circ}$$

$$AC = \frac{200 \cdot 0.743}{0.921} \approx 161.3 \text{ m}$$

Answer: The distance across the lake is approximately 161.3 m.

Application 2: Navigation

Problem: A ship leaves port and sails 40 km on a bearing of 60° . It then changes course and sails 30 km on a bearing of 150° . How far is the ship from the port?

Solution:

The angle between the two paths is:

$$150^\circ - 60^\circ = 90^\circ$$

Wait, we need to reconsider. The bearing changes from 60° to 150° , so the angle between the two paths is:

$$\theta = 150^\circ - 60^\circ = 90^\circ$$

Using the Law of Cosines:

$$d^2 = 40^2 + 30^2 - 2(40)(30) \cos 90^\circ$$

$$d^2 = 1600 + 900 - 0 = 2500$$

$$d = 50 \text{ km}$$

Answer: The ship is 50 km from the port.

16.8 Practice Exercises (Bài tập Luyện tập)

Exercise 16.1: Find the exact values of $\sin 150^\circ$, $\cos 150^\circ$, and $\tan 150^\circ$.

Exercise 16.2: In triangle ABC, $a = 9$, $b = 12$, and $\angle C = 60^\circ$. Find side c .

Exercise 16.3: In triangle ABC, $\angle A = 40^\circ$, $\angle B = 70^\circ$, and $c = 15$. Find sides a and b .

Exercise 16.4: In triangle ABC, $a = 8$, $b = 10$, $c = 12$. Find all three angles.

Exercise 16.5: In triangle ABC, $AB = 7$, $AC = 9$, and $\angle A = 45^\circ$. Find the area of the triangle.

16.9 Chapter Summary (Tóm tắt Chương)

Key Concepts

1. Extended Trigonometric Ratios:

- Defined for angles from 0° to 180°

- Supplementary angles:

$$\sin(180^\circ - \alpha) = \sin \alpha$$

,

$$\cos(180^\circ - \alpha) = -\cos \alpha$$

2. Law of Cosines:

- $a^2 = b^2 + c^2 - 2bc \cos A$

- Use for SSS or SAS cases
- Generalizes the Pythagorean theorem

3. Law of Sines:

- $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$

- Use for ASA, AAS, or SSA cases

4. Area Formulas:

- SAS:

$$S = \frac{1}{2}ab \sin C$$

- Heron:

$$S = \sqrt{p(p-a)(p-b)(p-c)}$$

Important Formulas Summary

Formula	When to Use
$a^2 = b^2 + c^2 - 2bccos A$	SSS or SAS
$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$	ASA, AAS, SSA
$S = \frac{1}{2}absin C$	Two sides and included angle
$S = \sqrt{p(p-a)(p-b)(p-c)}$	Three sides known

Problem-Solving Strategy

For solving triangles:

1. Identify what is given (SSS, SAS, ASA, AAS, SSA)
2. Choose the appropriate law (Cosines or Sines)
3. Find the unknown elements systematically
4. Check that angles sum to 180°

For area problems:

1. If two sides and included angle are known $\rightarrow$ use
$$S = \frac{1}{2}absin C$$
2. If three sides are known $\rightarrow$ use Heron's formula
3. If base and height are known $\rightarrow$ use
$$S = \frac{1}{2} \times \text{base} \times \text{height}$$

16.10 Review Exercises (Bài tập Ôn tập)

Review 16.1: Simplify without using a calculator:

$$(2 \sin 30^\circ + \cos 135^\circ - 3 \tan 150^\circ)(\cos 180^\circ - \cot 60^\circ)$$

Review 16.2: In triangle ABC, $AB = 8$, $BC = 10$, and $AC = 12$. Find:

- (a) The largest angle
- (b) The area of the triangle

Review 16.3: In triangle ABC, $\angle A = 35^\circ$, $\angle C = 85^\circ$, and $b = 20$. Find sides a and c .

Review 16.4: A tower stands on level ground. From point A, the angle of elevation to the top is 30° . From point B, 50 m closer to the tower, the angle of elevation is 45° . Find the height of the tower.

Review 16.5: Two ships leave a port at the same time. One sails at 15 km/h on a bearing of 40° , the other at 20 km/h on a bearing of 130° . How far apart are they after 2 hours?

Memory Aids (Gợi nhớ)

Law of Cosines mnemonic:

- "**C**osines for **C**orners" — use when you know two sides and the **included angle** (corner)
- Or when you know all three sides (SSS)

Law of Sines mnemonic:

- "**S**ines for **S**traight ratios" — relates sides to their opposite angles
- Use when you have angle-side pairs

Supplementary angles:

- **S**ame **S**ines (sin stays the same)
- **C**osines **C**hange sign (cos becomes negative)

Area formulas:

- "Half the product times the sine" →
$$S = \frac{1}{2}ab\sin C$$
- "Heron needs all sides" → Heron's formula needs a, b, c

Common mistakes to avoid:

- Don't use Law of Sines for SSS or SAS cases (use Law of Cosines instead)
- Remember that
$$\sin(180^\circ - \alpha) = \sin \alpha$$

(not negative!)

- When using Law of Cosines, don't forget the $-2bccos A$ term
 - Check that your angles sum to 180°
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End of Chapter 16